LiFE Makers EGYPT

Samsung Innovation Campus

SMART WATER
DISTRIBUTION & MONITORING SYSTEM

Samsung Innovation Campus | IoT Graduation Project

SMART CONTROL • FAIR DISTRIBUTION • SUSTAINABLE WATER



PREPARED BY

Ahmed Emad Atris
Abdallah Sherif Saeed
Omar Mohamed Hassan
Ahmed Gamal Hegab

REAL-TIME MONITORING
Measure water levels, flow rates and environmental conditions

AUTOMATED DISTRIBUTION
Smart control based on real demand across all zones

FAIR & EFFICIENT USAGE
Maintain supply for all sectors without interruption

SUSTAINABLE IMPACT
Better resource management for agriculture, residential and industrial needs

BUILDING A SMARTER AND MORE RESILIENT WATER FUTURE

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1. Abstract & Problem Statement

1.1 Abstract

Water is an essential resource for agriculture, residential communities, and industrial operations. However, managing its distribution across multiple demand zones can be challenging when monitoring is limited, control is performed manually, or abnormal operating conditions are not detected promptly. These challenges highlight the need for a connected system that improves visibility, supports timely decisions, and enables more organized water distribution.

Smart Water Distribution & Monitoring System is an IoT-based solution designed to monitor and manage water distribution across three zones: **Farm, Residential, and Industrial**. The proposed architecture integrates ESP32-based monitoring nodes with MQTT communication through HiveMQ Cloud, a Node-RED decision and control layer hosted on a Raspberry Pi, and a ThingsBoard dashboard for remote monitoring and interaction.

The system is designed to collect sensor data, process incoming measurements, apply configurable control logic, and issue valve-control commands according to the selected operating mode. It supports automatic and manual operation, valve control, abnormal-condition detection, and alert notifications. MQTT provides communication between the distributed devices and the central control layer, while the dashboard presents system information in an accessible visual interface.

The project aims to demonstrate how Internet of Things technologies can support more responsive, observable, and coordinated water-distribution management. Its architecture also provides a foundation for future extensions, including additional monitoring points, improved fault diagnosis, and data-driven optimization. The current prototype and software integration are evaluated through documented communication, control, and alarm tests, with hardware-level performance assessed separately according to the tests completed.

1.2 Problem Statement

The Challenge

Water distribution systems serving different types of users must respond to varying demand while maintaining reliable operation. Agricultural irrigation, residential consumption, and industrial processes may have different water requirements and operating priorities. Managing these zones without a unified monitoring and control mechanism can make it difficult to identify abnormal conditions, understand the current state of the system, and respond consistently to changing requirements.

Several challenges motivate this project:

- **Limited real-time visibility:** Without a centralized monitoring interface, obtaining an overall view of water-distribution conditions across multiple zones can be difficult.
- **Delayed detection of abnormal conditions:** Unexpected readings, abnormal flow behavior, or sensor communication failures may remain unnoticed unless monitoring and alarm mechanisms are available.
- **Fragmented control:** Managing multiple distribution valves independently can complicate coordination between zones and make operating status harder to track.
- **Dependence on manual intervention:** Manual monitoring and control can increase the effort required to supervise a distributed system and respond to changing conditions.
- **Lack of integrated information:** Sensor measurements, valve commands, operating modes, and alarm notifications are more useful when brought together within a common monitoring and control architecture.

The Proposed Approach

This project addresses these challenges through an integrated IoT architecture that connects monitoring nodes, a message broker, a central decision layer, controllable valves, and a unified dashboard. The system is designed to process incoming measurements, apply defined control rules, support automatic and manual operating modes, and notify the operator when configured abnormal conditions are detected.

Rather than treating monitoring and control as separate functions, the proposed solution connects them in a common workflow: **measure, communicate, evaluate, act, and monitor**. This approach is intended to improve operational visibility and provide a structured basis for managing water distribution across the three designated zones.

The project focuses on demonstrating the feasibility of this architecture through its implemented software components, documented tests, and prototype or simulation evidence. Claims about water savings, distribution efficiency, or long-term reliability require further measurement and validation under representative operating conditions.

1.3 Evidence of the Real-World Problem

Evidence 01: Irrigation Water Disruption in Abu Suwir, Ismailia:



Source: Al-Dostor Newspaper
Publication Date: April 28, 2026
Article: End of the irrigation water interruption crisis in Abu Suwir and the return of water to agricultural lands

Evidence Description

A news report published by Al-Dostor documented complaints about irrigation water interruptions affecting several agricultural areas in Abu Suwir, Ismailia Governorate. According to the report, the interruption occurred after the closure of the main irrigation channel supplying these areas during maintenance and cleaning works on the Ismailia Canal. The incident affected agricultural land and crops, prompting local authorities to intervene and address the problem.

Relevance to Our Project

This incident demonstrates how interruptions affecting a main water supply route can disrupt water delivery to the agricultural areas that depend on it. It highlights the importance of monitoring water distribution, identifying supply interruptions, and managing water delivery across multiple zones.

Our Smart Water Distribution and Monitoring System addresses this general challenge through sensor-based monitoring, centralized decision-making, and adjustable valves for controlling water delivery to different zones.

Evidence Limitation

This report documents an interruption associated with maintenance works. It does not establish that unequal water allocation or excessive consumption caused the incident. Therefore, it supports the need for reliable water distribution and monitoring, but it is not direct evidence of demand-based water shortages.

Evidence 02 — Drinking Water Interruption in Ismailia:



Source: Al-Masry Al-Youm Newspaper
Publication Date: August 1, 2026
Article: Drinking Water Interruptions: A Recurring Concern for Ismailia Residents

Evidence Description

A report published by Al-Masry Al-Youm documented repeated drinking water interruptions affecting several residential areas in Ismailia Governorate. The report described residents experiencing prolonged service disruptions, including complaints of a complete water interruption lasting several days in the El-Kilo 2 area. According to the report, dredging activities and increased water turbidity affected the operation of water treatment stations.

Relevance to Our Project

This case highlights the impact that disruptions in water infrastructure can have on residential users. It supports the need for continuous monitoring, timely identification of abnormal operating conditions, and better visibility of water system status. Our project applies these principles by collecting sensor data, processing operating conditions through a centralized Node-RED flow, and providing dashboard monitoring and alarm notifications.

Evidence Limitation

The reported interruption was associated with water quality and maintenance activities. The article does not establish that the incident was caused by excessive demand or unfair water allocation between residential, agricultural, and industrial users.

Evidence 03: Water Treatment Station Shutdown in Ismailia:



Source: Youm7 Newspaper
Publication Date: July 19, 2026
Article: Increased Turbidity in the Ismailia Canal Temporarily Shuts Down Two Water Stations

Evidence Description

A report published by Youm7 stated that two drinking water treatment stations in Ismailia, located at Kilometer 7 and Kilometer 11, were temporarily shut down because of increased water turbidity associated with dredging and cleaning works in the Ismailia Canal. The report identified the residential areas affected by the shutdown of each station. The temporary suspension was intended to protect water quality and public safety.

Relevance to Our Project

This incident highlights that water distribution depends not only on controlling valves and allocating water, but also on monitoring the condition of the water source and the operational status of the infrastructure. It supports the importance of timely monitoring, centralized status visibility, and alert mechanisms that help operators identify abnormal conditions and respond appropriately.

Our project demonstrates a monitoring and control architecture that collects sensor readings, processes operating conditions centrally through Node-RED, and presents system status and alarms through a dashboard.

Evidence Limitation

The reported shutdown was a precautionary response to increased turbidity during maintenance works. It does not prove that excessive demand or unequal allocation caused the interruption. It is included as evidence of the broader operational challenges involved in maintaining a reliable water supply.

Evidence Summary

The reviewed news reports document real-world water supply disruptions affecting agricultural areas, residential communities, and drinking water treatment stations in Ismailia Governorate. These incidents illustrate how interruptions in water infrastructure and changes in source-water conditions can affect the availability and continuity of water services.

Together, the evidence supports the need for improved monitoring, timely identification of abnormal conditions, and coordinated visibility across different water distribution zones. These needs are directly relevant to the proposed Smart Water Distribution and Monitoring System, which integrates sensor data, centralized decision-making through Node-RED, controllable valves, and dashboard-based monitoring and alerts.

However, the reviewed reports do not establish that excessive demand or unequal allocation was the direct cause of the documented incidents. They also do not demonstrate measurable water savings achieved by our prototype. Such outcomes would require dedicated testing and quantitative validation.

1.4 Project Objectives

The main objective of this project is to develop an IoT-based water distribution and monitoring system that demonstrates how water can be managed across agricultural, residential, and industrial zones using sensor data, centralized control, and adjustable valves.

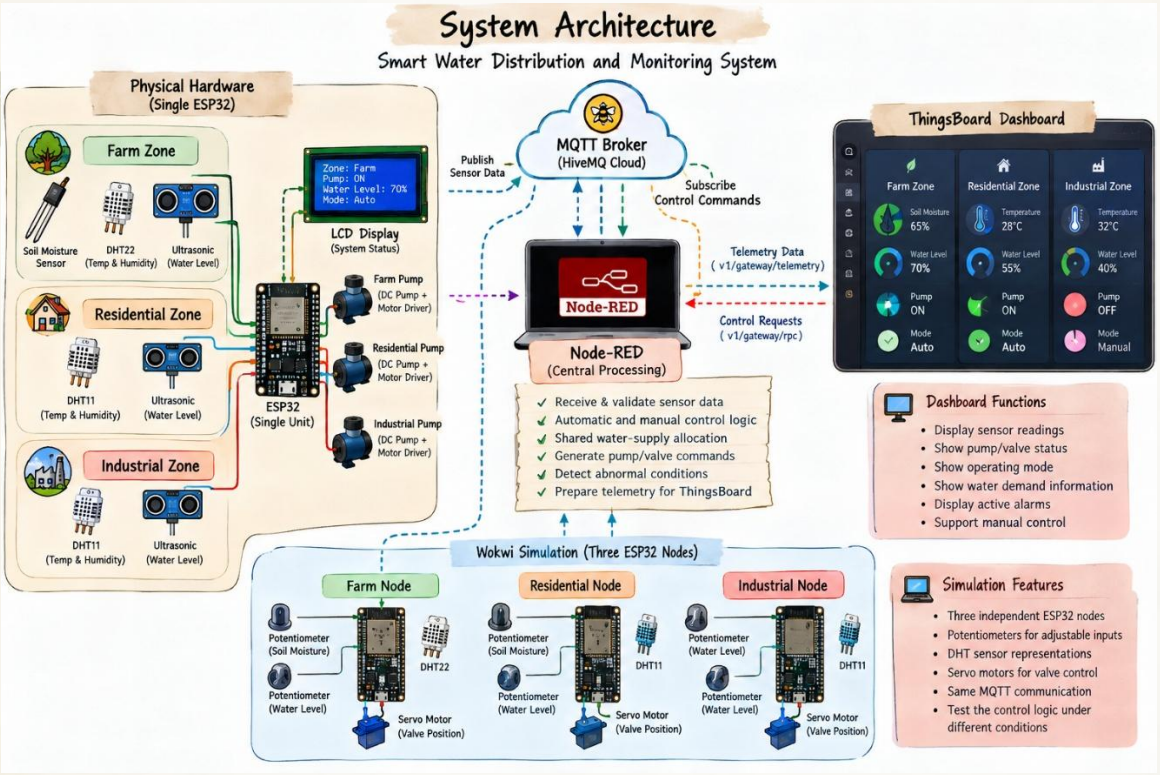
The specific objectives are:

- **Monitor Water Conditions Across Multiple Zones:** Collect sensor readings to provide visibility into the operating conditions of the agricultural, residential, and industrial branches.
- **Respond to Varying Water Requirements:** Consider that each zone has different water needs depending on its purpose and operating conditions, rather than treating all zones as having identical requirements.
- **Implement Adjustable Valve Control:** Use controllable valves to regulate water delivery to each branch at different opening levels instead of relying only on fully open or fully closed states.
- **Support Data-Driven Agricultural Irrigation:** Use soil moisture, temperature, humidity, water-level, and rainfall-related information, where available in the prototype, to support irrigation decisions.
- **Centralize Monitoring and Control Logic:** Use Node-RED as the central decision-making layer to process incoming data, apply configured conditions, coordinate valve commands, and monitor system status.
- **Enable Communication Between System Components:** Use ESP32-based nodes and MQTT through HiveMQ Cloud to exchange sensor readings and control commands between the distributed nodes and the central monitoring system.
- **Provide a Unified Monitoring Interface:** Present relevant measurements, operating modes, valve states, and alerts through a centralized ThingsBoard dashboard.
- **Evaluate the Prototype Under Different Operating Conditions:** Test how the system responds to changing sensor readings and zone requirements, and document its behavior to assess the feasibility of the proposed approach.

The project aims to demonstrate a structured approach to monitoring and controlling water distribution across three representative zones. Fair allocation and uninterrupted supply are design goals; their achievement and any measurable reduction in water consumption require further testing and quantitative validation.

1.5 System Architecture

The Smart Water Distribution and Monitoring System monitors and controls water distribution across three zones: Farm, Residential, and Industrial. The project combines physical hardware, Wokwi simulation, MQTT communication through HiveMQ Cloud, centralized processing using Node-RED, and a ThingsBoard dashboard.



1.5.1 Physical Hardware Architecture

The physical prototype uses a single ESP32 to monitor the Farm, Residential, and Industrial zones. The Farm zone includes a soil moisture sensor, a DHT22 temperature and humidity sensor, and an ultrasonic sensor for water-level measurement. The Residential and Industrial zones use DHT11 temperature and humidity sensors and ultrasonic water-level sensors.

The ESP32 communicates with the central control system through MQTT. Water is pumped using DC pumps controlled through motor-driver circuits. Pump operation receives commands from the current Node-RED system, allowing the physical setup to respond to centralized control decisions.

An LCD will be added to the physical prototype to display the system's local operating status. It will provide a quick view of relevant information, such as the active zone, pump status, water level, and operating mode, depending on the available data. This will allow users to check the system status directly at the hardware level without relying entirely on the remote ThingsBoard dashboard.

1.5.2 Simulation Architecture

The Wokwi simulation uses three separate ESP32 nodes, one for each zone, communicating through the same MQTT broker. This arrangement allows the zones to be tested independently while following the same centralized monitoring and control approach.

Potentiometers provide adjustable inputs to represent variable sensor readings, including soil moisture and water level. Temperature and humidity sensors are also represented in the simulation. Servo motors are used to represent adjustable valve positions, allowing the valve-control logic to be tested in simulation, while the physical prototype uses DC pumps for water movement rather than servo-operated valves.

1.5.3 MQTT Communication and HiveMQ Cloud

MQTT provides communication between the ESP32 nodes and Node-RED through HiveMQ Cloud. Each zone has dedicated topics for sensor readings, valve commands, and operating modes:

- Sensor data: smartwater/<zone>/sensors
- Valve commands: smartwater/<zone>/valve
- Operating mode: smartwater/<zone>/mode

The <zone> identifier is farm, residential, or industrial. This structure keeps the data and commands for each zone separate while allowing the central system to coordinate all three zones.

1.5.4 Central Processing Using Node-RED

Node-RED acts as the central decision-making and control layer. It receives and validates sensor data from the three zones, applies the configured decision logic, and manages automatic and manual operating modes.

The system coordinates shared water-supply allocation, generates zone-specific valve commands for the simulation, and processes pump-control commands for the physical hardware where applicable. Node-RED also monitors configured abnormal conditions and prepares telemetry and alarm information for integration with ThingsBoard. This architecture keeps the central control logic separate from the local hardware execution performed by the ESP32.

1.5.5 Monitoring Interface Using ThingsBoard

ThingsBoard provides a dashboard for monitoring the three zones. The available telemetry includes sensor readings, operating modes, valve positions, water-demand information, and alarm status. The integration also supports the dashboard control requests implemented in the project.

This interface provides a common view of the system and allows the operator to monitor zone conditions and use the supported control functions.

1.5.6 System Operation Overview

The system follows a continuous monitoring and control cycle. Each ESP32 collects sensor readings and publishes them to its zone-specific MQTT topic. Node-RED receives and validates the data, applies the configured control logic, and publishes the appropriate valve commands. Operating information and telemetry are then forwarded to ThingsBoard for monitoring.

The physical prototype uses one shared ESP32, whereas the Wokwi simulation uses three ESP32 nodes. Both configurations follow the same general communication and centralized-control concept, while their hardware arrangements differ.

1.6 Software Design and Control Logic

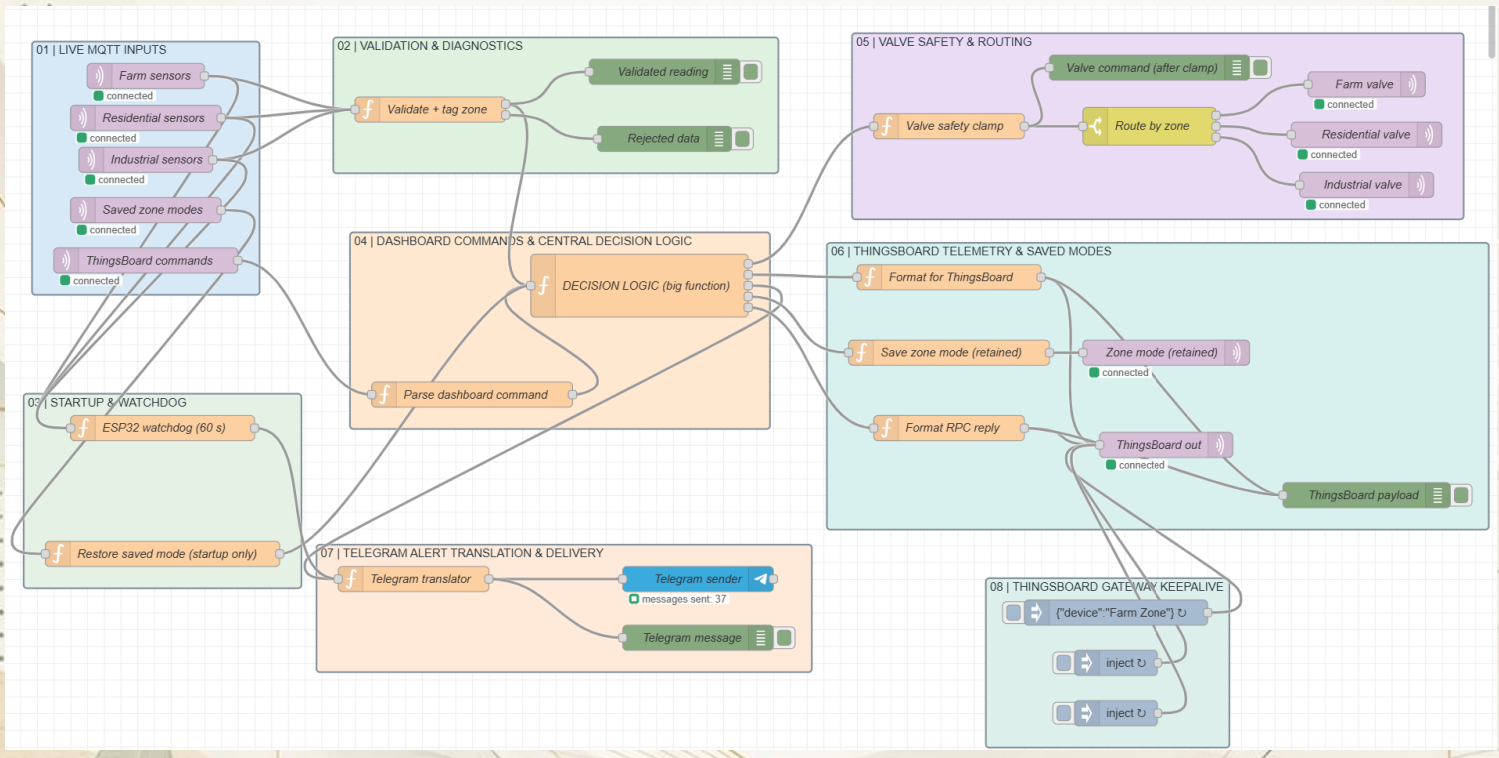
The software design connects sensor acquisition, centralized decision-making, hardware control, and remote monitoring. The ESP32 handles sensor readings and executes received commands, while Node-RED processes the data and coordinates the system operation. HiveMQ Cloud provides MQTT communication, and ThingsBoard presents the system status through a centralized dashboard.

1.6.1 Software Architecture

The software consists of four main components:

- **ESP32 Microcontroller:** Collects sensor readings, publishes data through MQTT, receives commands, and executes them through the connected hardware.
- **HiveMQ Cloud:** Provides message exchange between the ESP32 nodes and Node-RED through zone-specific MQTT topics.
- **Node-RED:** Validates sensor data, applies the configured decision logic, manages operating modes and shared water-supply allocation, and generates control commands and alarm information.
- **ThingsBoard:** Displays sensor measurements, valve positions, operating modes, water-demand information, and alarm status, while supporting the implemented dashboard control functions.

This architecture separates local hardware execution from centralized decision-making and remote monitoring, making the system easier to manage across the three zones.



NODE RED CONTROL FLOW

1.6.2 MQTT Communication and Topic Structure

MQTT organizes communication between the ESP32 nodes and Node-RED through zone-specific topics hosted on HiveMQ Cloud. Each zone uses separate topics for sensor measurements, valve commands, and operating modes.

Sensor Data Topic

smartwater/<zone>/sensors

Each ESP32 publishes its zone's sensor readings to this topic. Node-RED receives the messages and validates the available measurements before using them in the control logic.

Valve Control Topic

smartwater/<zone>/valve

Node-RED publishes valve-control commands through the corresponding zone topic. In the Wokwi simulation, the ESP32 receives these commands and controls the associated servo position.

Operating Mode Topic

smartwater/<zone>/mode

This topic carries the operating mode for each zone. Retained messages allow the latest published mode to remain available to new subscribers.

The <zone> identifier is farm, residential, or industrial, keeping the communication for each zone logically separated.

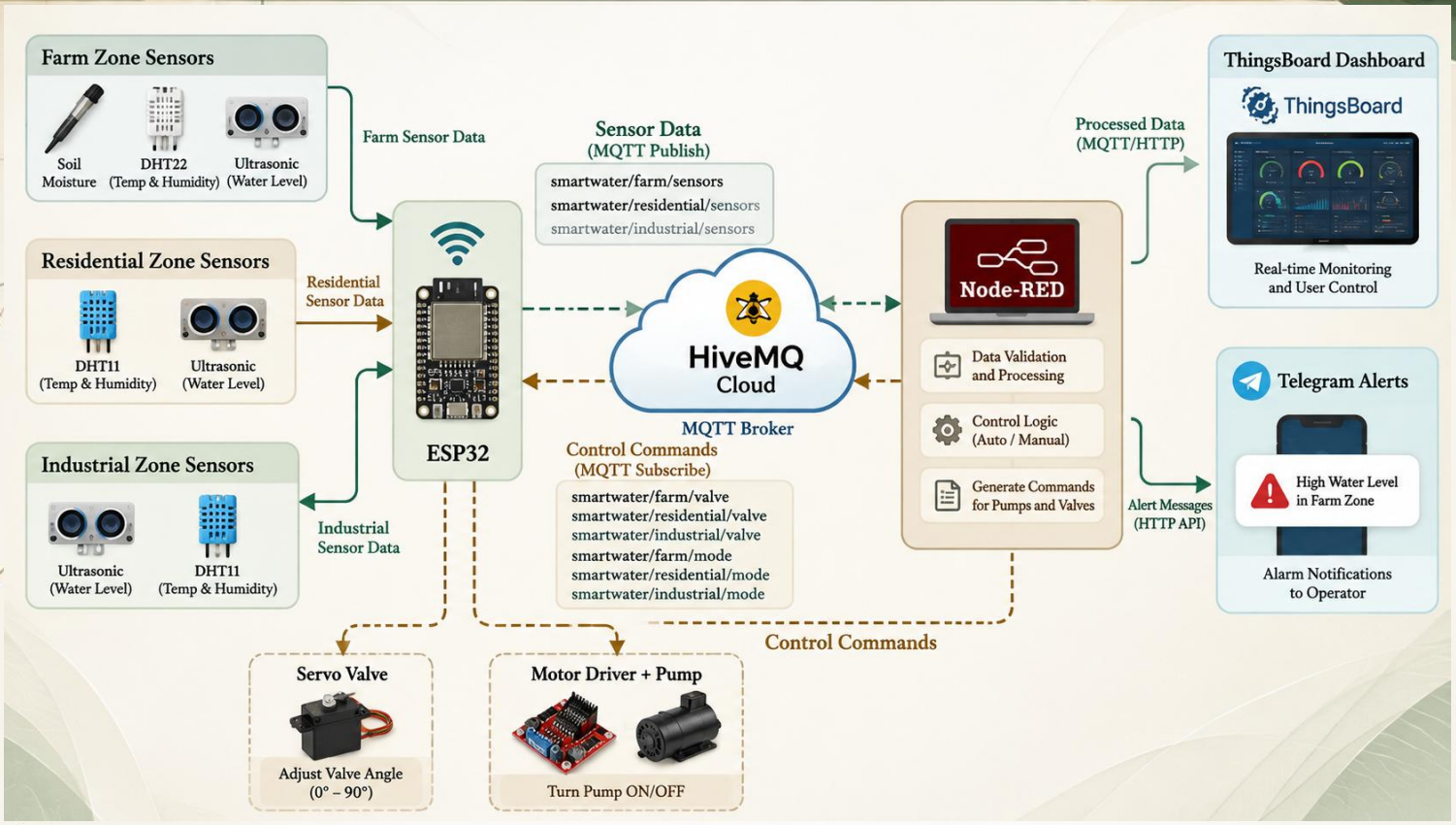
ThingsBoard Gateway Communication

Node-RED also communicates with ThingsBoard through its MQTT gateway interface. The main topics used by the current flow are:

- Telemetry: v1/gateway/telemetry
- Remote procedure calls (RPC): v1/gateway/rpc

Telemetry carries system measurements and operating information to the dashboard, while the RPC topic supports the implemented remote-control requests.

This structure separates sensor communication and valve commands from dashboard integration, while allowing Node-RED to coordinate the three zones centrally.



And This Figure illustrates the updated system architecture, showing how the physical hardware communicates with HiveMQ Cloud and Node-RED. Node-RED handles sensor validation, decision logic, shared water-supply allocation, operating modes, and alarm detection, while ThingsBoard provides centralized monitoring and control. The physical prototype uses DC pumps, whereas servo valves are used to represent valve control in the Wokwi simulation.

1.6.3 Automatic and Manual Control Logic

The system supports three operating states: Automatic, Manual, and Closed. In Automatic mode, Node-RED evaluates sensor readings and zone demand to determine the required valve position. The control process considers water-level conditions, environmental measurements, and available flow feedback to adjust the valve angle according to the configured control parameters.

A shared water-supply allocation mechanism coordinates the Farm, Residential, and Industrial zones. It calculates demand and target flow, then considers the configured minimum supply shares and zone priorities when distributing the available water supply.

In Manual mode, valve-position commands can be applied through the supported control interface, subject to the configured command limits. In Closed mode, the valve command is set to 0°. The current control flow also includes timing constraints and a deadband to avoid unnecessary valve adjustments.

The system monitors sensor validity and operating conditions while applying the control logic. Detected abnormal conditions are used to generate alarm information for monitoring and notification.

1.6.4 Monitoring and Alarm Management

Node-RED continuously monitors incoming sensor data and zone operating conditions to identify abnormal situations. The current flow includes alarms for low or high water levels, abnormal flow readings, possible leakage while a valve is closed, invalid sensor readings, and ESP32 communication loss.

Additional alarms cover zones operating in the closed or manual state, multiple zones being closed simultaneously, rejected control commands, and system startup. Sensor faults are detected after consecutive invalid readings, while an ESP32 is considered offline if no sensor update is received for more than 60 seconds.

The generated alarm information is forwarded to the monitoring system and prepared for Telegram notifications. ThingsBoard displays available alarm information and zone status, helping the operator identify conditions that may require attention.

1.6.5 System Verification and Test Cases

The system was verified by publishing test sensor readings through HiveMQ Cloud and observing the resulting behavior in Node-RED and ThingsBoard. Each test was used to check whether the system correctly validated the input, identified the corresponding condition, and updated the reported alarm status.

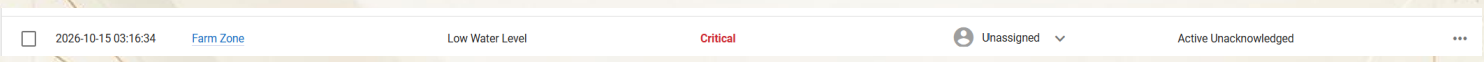
Test Case 1: Low Water Level Detection

Input Topic: smartwater/farm/sensors

Input Payload: { "rain": 0, "temp": 25, "level": 5, "flow": 10}

Expected Behavior: The system should classify the Farm water level as low and activate the LEVEL_LOW alarm.

Observed Result: Node-RED successfully validated the incoming readings. The ThingsBoard telemetry showed levelBand: low, activeAlarms: LEVEL_LOW, and alarmCount: 1. The reported valve angle was 90°, the calculated demand was 98%, and no sensor fault was reported.



Status: Passed.

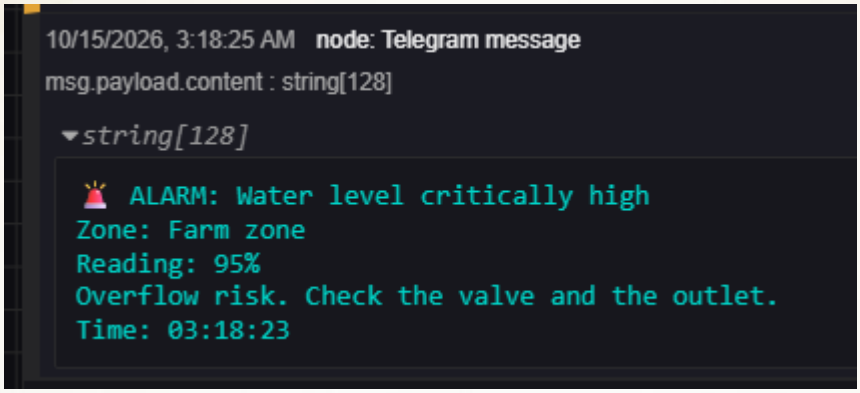
Test Case 2: High Water Level Detection

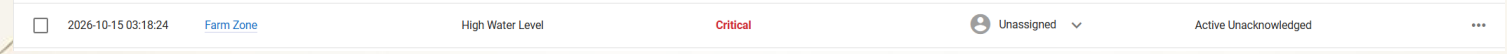
Input Topic: smartwater/farm/sensors

Input Payload: { "rain": 0, "temp": 25, "level": 95, "flow": 10 }

Expected Behavior: The system should classify the Farm water level as high and activate the LEVEL_HIGH alarm.

Observed Result: Node-RED detected the critically high water level and generated an alert at 95%. ThingsBoard displayed an active, unacknowledged High Water Level alarm with Critical severity for the Farm Zone.





Status: Passed.

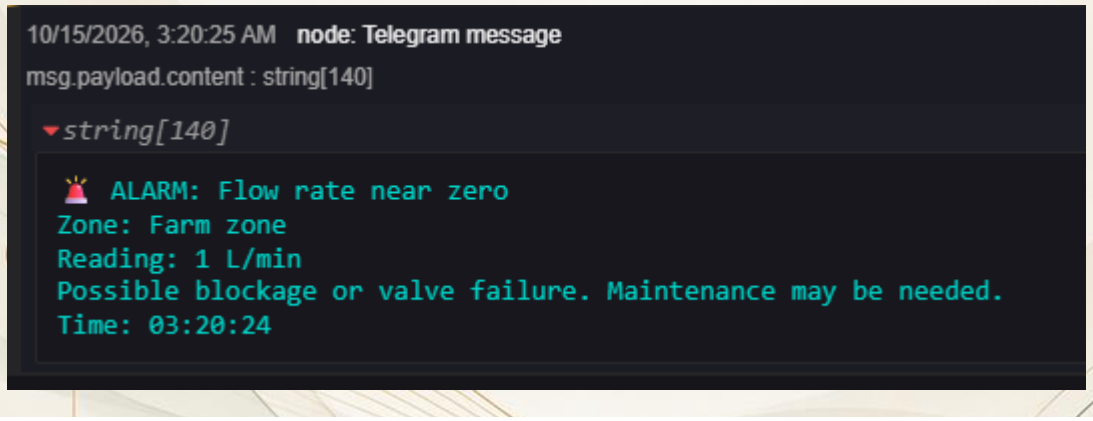
Test Case 3: Abnormal Flow Detection

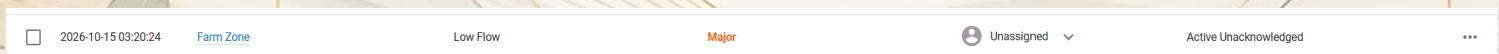
Input Topic: smartwater/farm/sensors

Input Payload: { "rain": 0, "temp": 25, "level": 50, "flow": 1 }

Expected Behavior: The system should detect abnormally low water flow and activate the corresponding low-flow alarm.

Observed Result: Node-RED detected a near-zero flow rate of 1 L/min and reported a possible blockage or valve failure. ThingsBoard successfully displayed an active, unacknowledged Low Flow alarm with Major severity for the Farm Zone.





Status: Passed.

Test Case 4: High Temperature Detection

Input Topic: smartwater/farm/sensors

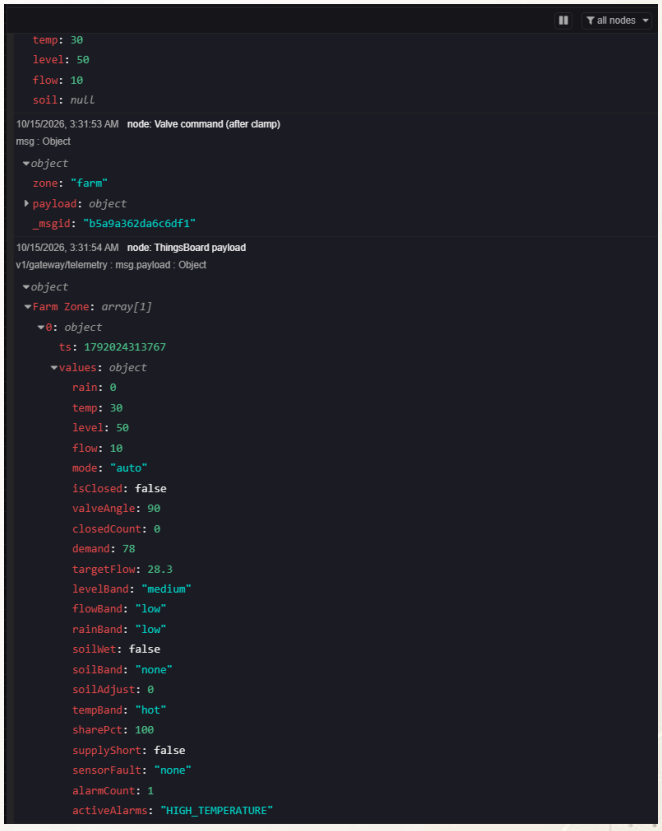
Input Payload: { "rain": 0, "temp": 30, "level": 50, "flow": 10 }

Expected Behavior: The system should detect high temperature in the Farm Zone when the temperature exceeds 25°C and activate the corresponding High Temperature alarm.

Observed Result: Node-RED detected the high temperature at 30°C and classified it as hot. The HIGH_TEMPERATURE alarm was activated, and alarmCount increased to 1. The sensorFault status remained none, confirming that the temperature alert was triggered without a sensor fault.



Status: Passed.



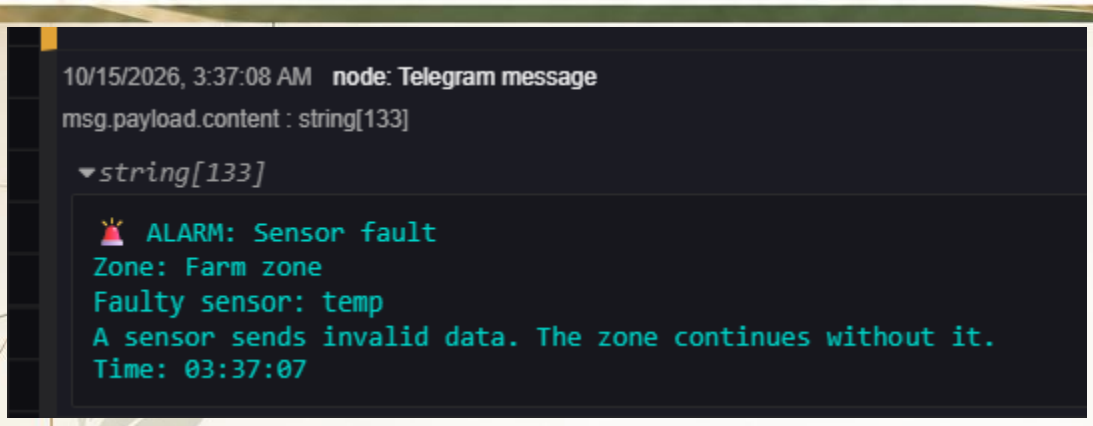
Test Case 5: Sensor Fault Detection

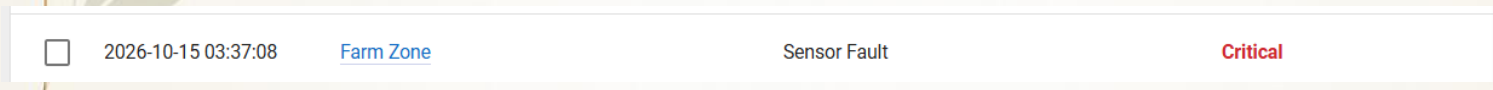
Input Topic: smartwater/farm/sensors

Input Payload: { "rain": 0, "temp": 25, "level": 50, "flow": 10, "soil": null }

Expected Behavior: The system should detect invalid sensor readings after three consecutive failures and activate the SENSOR_FAULT alarm while allowing the zone to continue operating.

Observed Result: Node-RED continued receiving and validating sensor readings, but no SENSOR_FAULT alarm was observed in the provided logs. The test result remains inconclusive and requires three consecutive invalid sensor readings to verify fault detection and alarm activation.





Status: Passed.

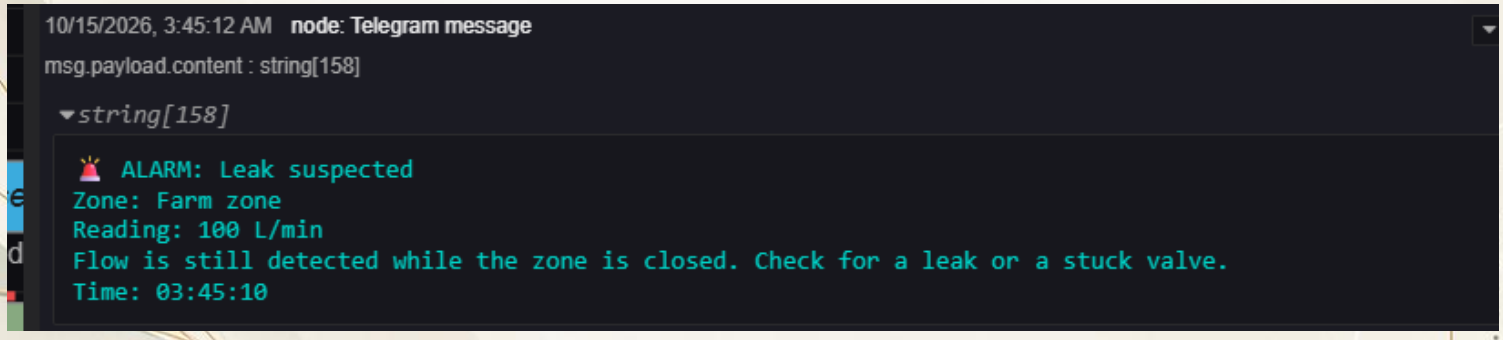
Test Case 6: Leak Detection

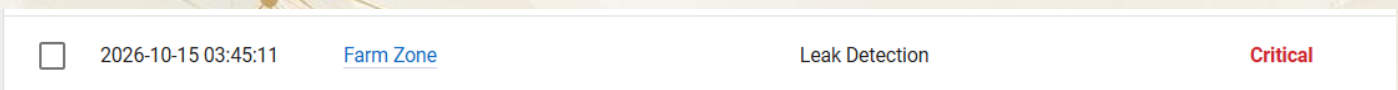
Input Topic: smartwater/farm/sensors

Input Payload: { "flow": 100, "isClosed": true, "mode": "closed" }

Expected Behavior: The system should detect continued water flow while the Farm Zone is closed for more than 30 seconds and activate the LEAK alarm.

Observed Result: Node-RED detected a flow rate of 100 L/min while the Farm Zone was closed and generated a Leak Suspected alert. The notification indicated a possible leak or stuck valve.



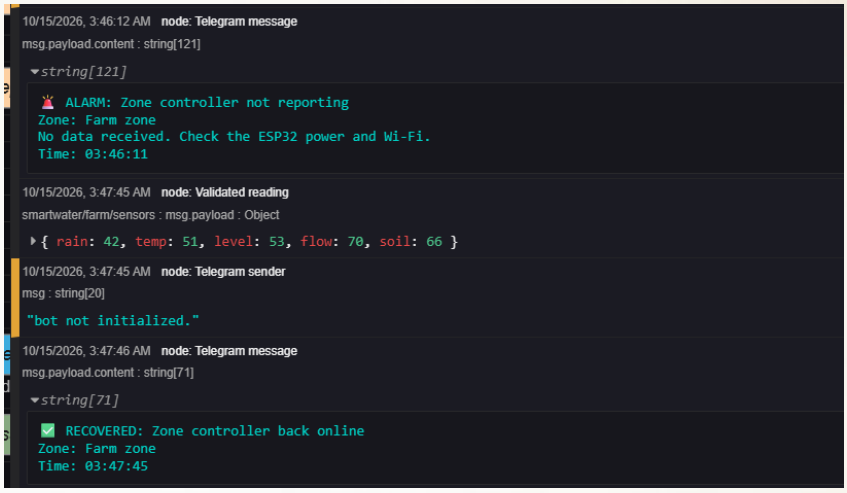


Status: Passed.

Test Case 7: ESP Offline & Recovery

Input Topic: smartwater/farm/sensors

Input Payload: No telemetry messages received from the Farm Zone ESP32 for more than 60 seconds.



Expected Behavior: The system should detect that the Farm Zone ESP32 is offline after 60 seconds without telemetry and activate the ESP_OFFLINE alarm. When the ESP32 reconnects and telemetry resumes, the alarm should be cleared.

Observed Result: Node-RED detected the Farm Zone controller outage and generated an offline alert. After telemetry resumed, the controller was reported back online, confirming successful detection of disconnection and recovery.

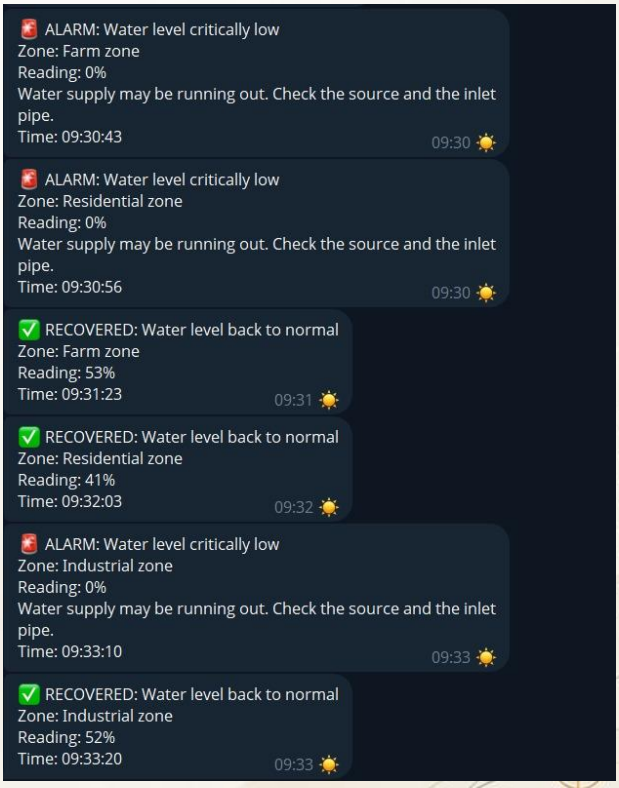
Test Case 8: Critical Water Level Detection and Recovery

Input Topic: smartwater/{zone}/sensors

Input Payload: Water level readings of 0%, followed by recovery readings of 53% for Farm, 41% for Residential, and 52% for Industrial.

Expected Result: The system should send a Telegram alarm when the water level becomes critically low and send a recovery notification when the level returns to a normal range. The notification should identify the affected zone and report the measured percentage.

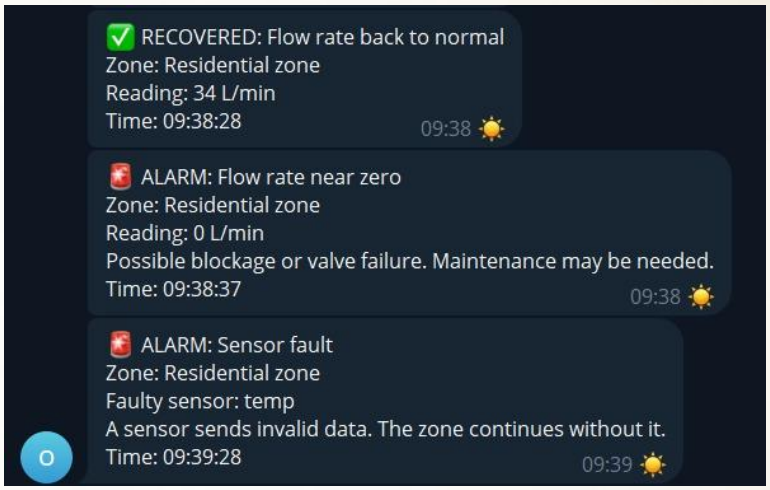
Observed Result: Telegram notifications confirmed critical low-water alarms at 0% for all three zones. Recovery notifications were also received, reporting 53% for Farm, 41% for Residential, and 52% for Industrial.



Test Case 9: Flow Anomaly and Sensor Fault Detection

Input Topic: smartwater/{zone}/sensors

Input Payload: Residential flow rate drops to 0 L/min and remains at zero for approximately 50 seconds.

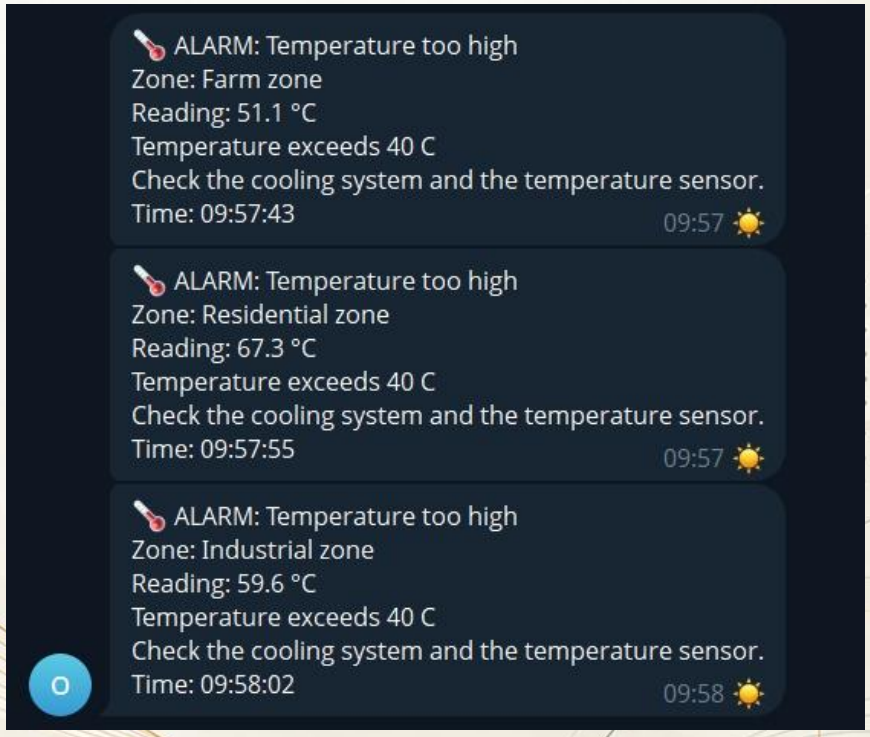


Expected Result: The system should detect an abnormal near-zero flow condition and notify the operator. If the condition persists for the configured detection period, a sensor-fault notification should be generated.

Observed Result: Telegram first reported that the Residential flow rate had returned to normal at 34 L/min. A subsequent alert reported flow at 0 L/min, indicating a possible blockage or valve failure. After approximately 50 seconds, a Sensor Fault notification identified the faulty sensor as temp. This confirms the reported fault-detection behavior, although the exact cause of the invalid reading requires further investigation.

Test Case 10: High Temperature Detection Across All Zones

Input Topic: smartwater/{zone}/sensors



Input Payload: Temperature readings above 40°C for Farm (51.1°C), Residential (67.3°C), and Industrial (59.6°C).

Expected Result: The system should send a Telegram alarm whenever a zone's temperature exceeds the configured 40°C threshold. Each notification should identify the affected zone, include the measured temperature, and advise the operator to inspect the cooling system and temperature sensor.

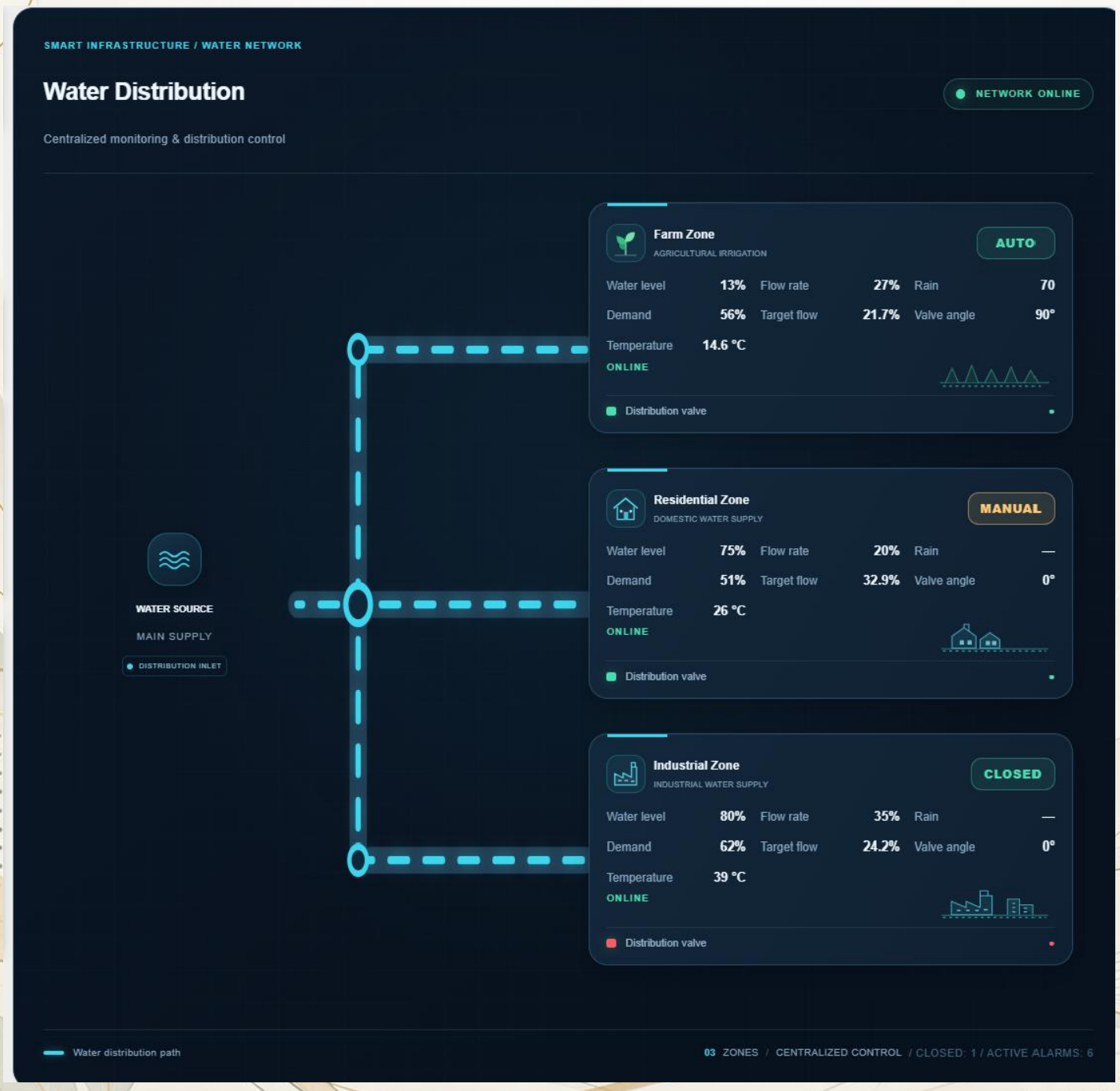
Observed Result: Three Telegram alarms were received successfully. The notifications reported 51.1°C for Farm, 67.3°C for Residential, and 59.6°C for Industrial, confirming that high-temperature alerts were delivered for all three zones.

2. Dashboard Design and Operation

2.1 Dashboard Overview

The dashboard was developed using ThingsBoard to provide a centralized interface for monitoring and controlling the Smart Water Distribution System. It includes three main zones: Farm, Residential, and Industrial. Each zone has its own monitoring panel, displaying its current measurements, operating mode, valve status, and available control options.

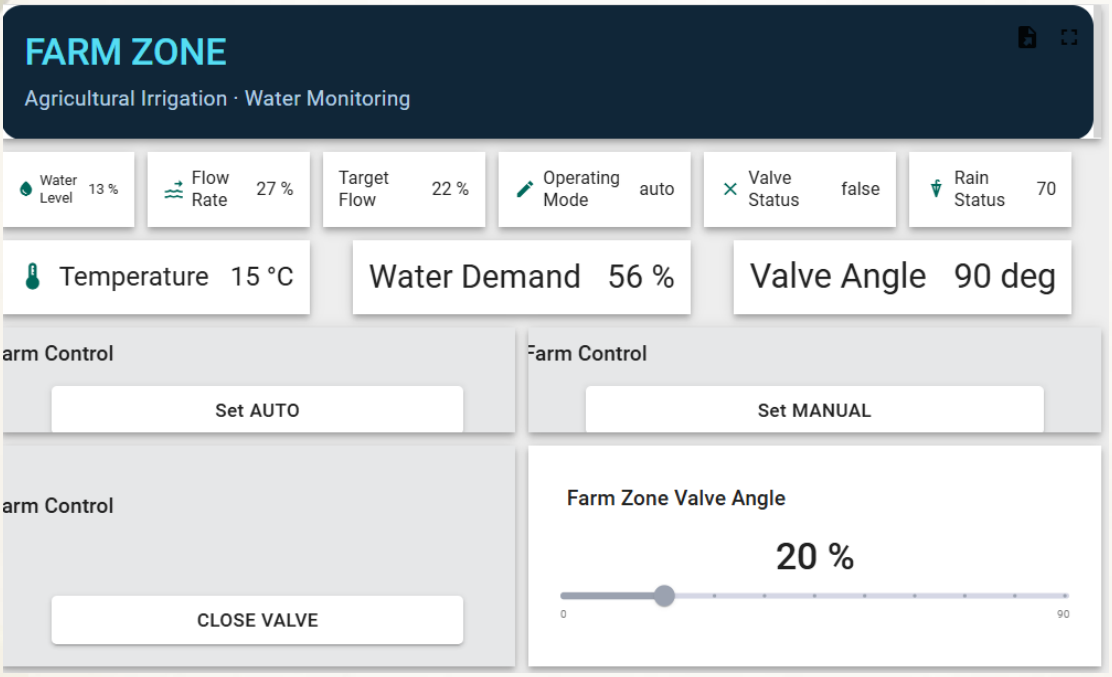
The dashboard also includes a centralized water distribution visualization that represents the three zones within the same network. This allows the operator to monitor the overall system while accessing the individual controls of each zone.

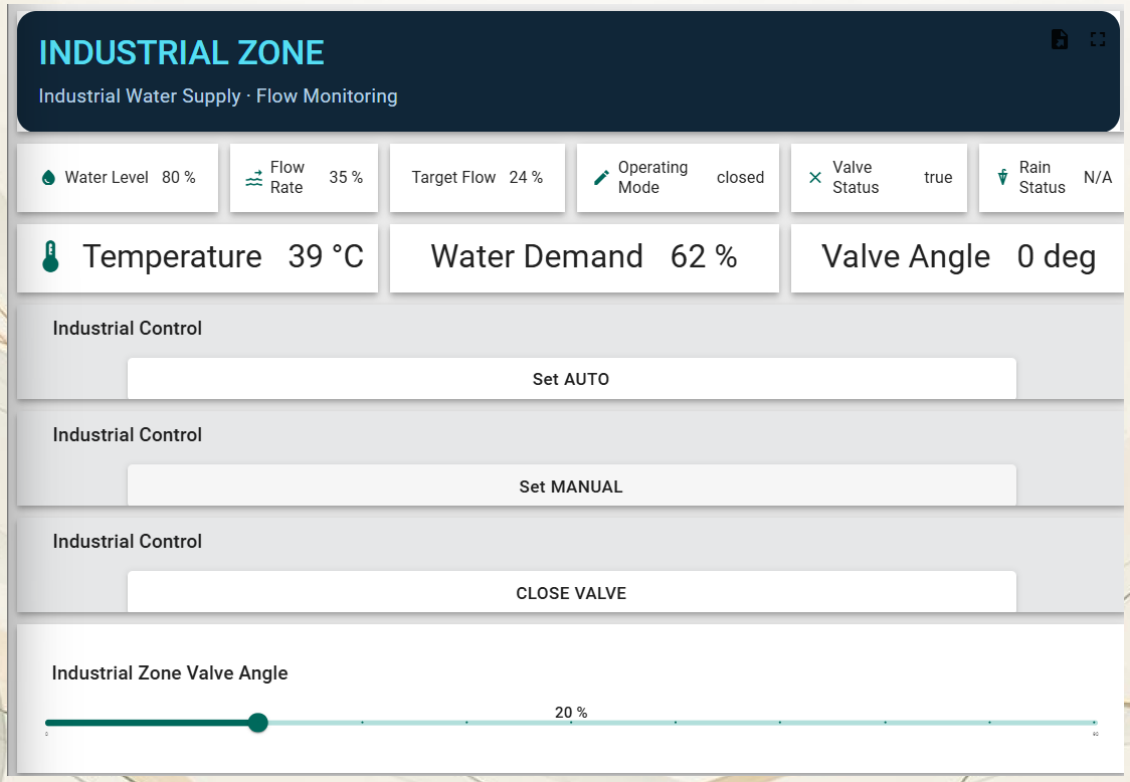


2.2 Monitoring and Displayed Parameters

Each zone panel displays the main operating parameters, including Water Level, Flow Rate, Target Flow, Temperature, Water Demand, Operating Mode, Valve Status, Valve Angle, and Rain Status where applicable.

These indicators help the operator understand the current condition of each zone. The displayed values are updated according to the telemetry received by ThingsBoard. The Farm Zone also supports agricultural monitoring, where environmental measurements are relevant to irrigation.

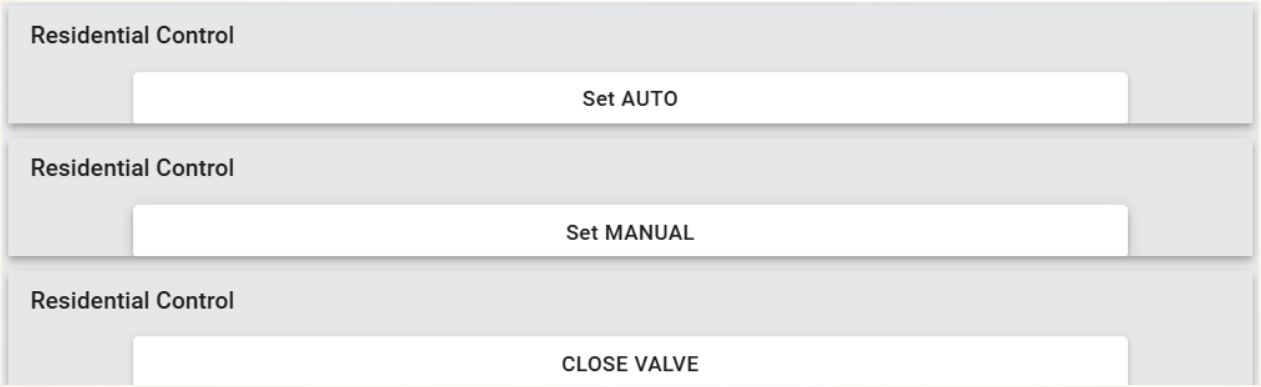




2.3 Automatic ,Manual and Closed Operating Modes

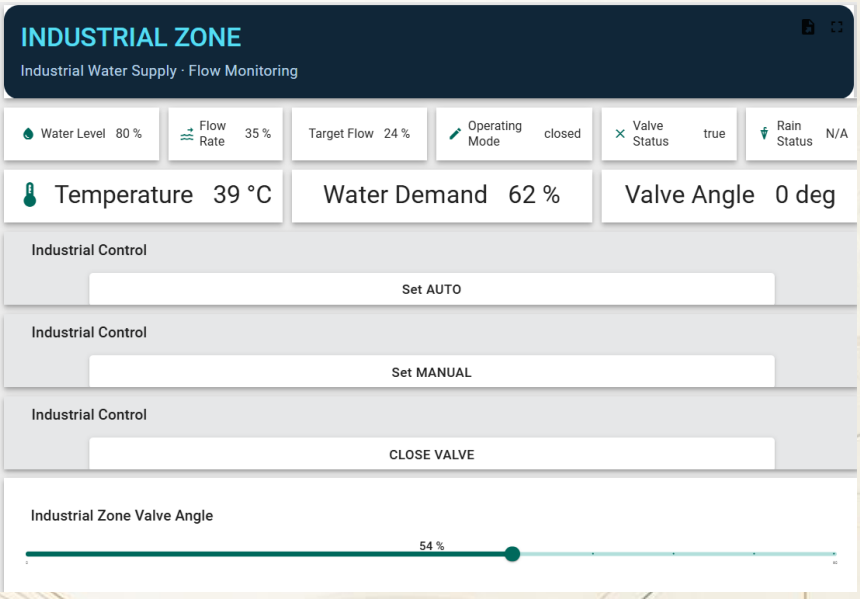
The dashboard provides three main operating modes: **AUTO**, **MANUAL** and **CLOSED.**

- **AUTO Mode:** The system uses its control logic to determine the required valve operation according to the monitored conditions and calculated water demand. The operator can monitor the resulting measurements and valve status through the dashboard.
- **MANUAL Mode:** The operator can take direct control of the zone's valve using the available control widgets. This mode allows the valve position to be adjusted manually when required.
- **CLOSED State:** The valve can be commanded to close using the CLOSE VALVE control. The displayed valve status and angle indicate the reported state of the valve.



2.4 Valve Control and Page Refresh Behavior

Each zone includes a CLOSE VALVE button and a valve-angle slider for controlling the valve through the dashboard. In MANUAL mode, the operator can adjust the requested valve angle using the slider, allowing the valve position to be changed according to the required operating condition.



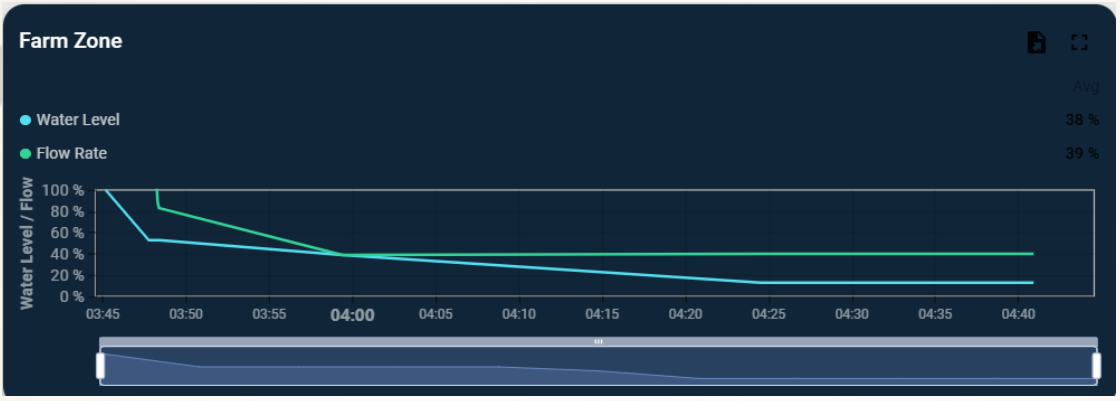
During operation, opening the valve in MANUAL mode may require refreshing the dashboard page before the updated valve state is displayed or the expected operation becomes visible. This behavior is a current limitation of the dashboard interaction and should be considered when operating the manual controls.

The AUTO and MANUAL controls remain available for switching between automatic operation and manual intervention. The dashboard also displays the reported valve status and angle so that the operator can observe the system response after issuing a command.

2.5 Historical Charts

Each zone has a time-series chart showing Water Level and Flow Rate over time. These charts help the operator observe variations in the monitored values and identify changes in water distribution conditions.

The charts become more informative as additional telemetry readings are received. They complement the numerical indicators by showing trends rather than only the latest measurements.







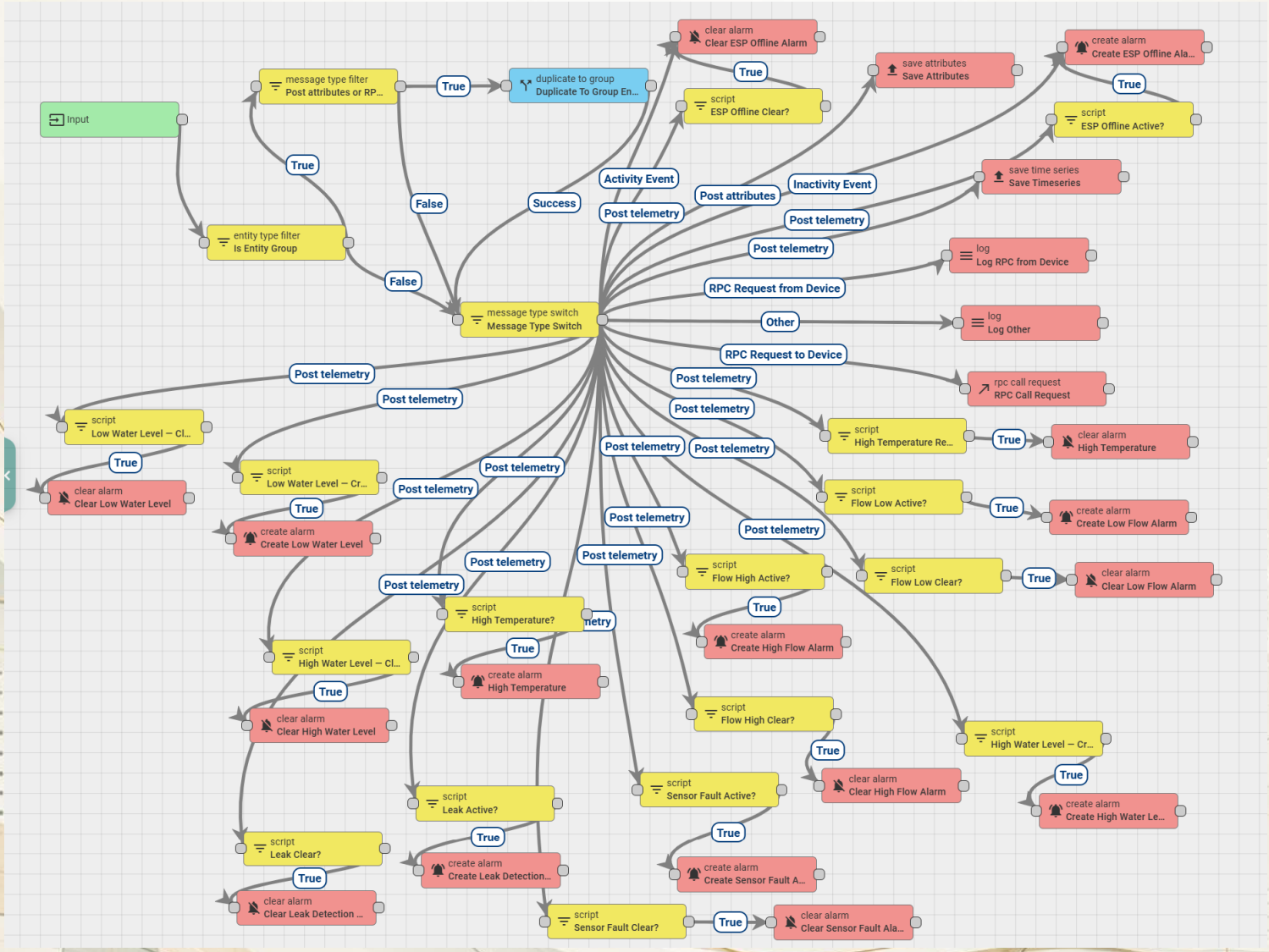
2.6 Centralized Network Visualization

The centralized Water Distribution view presents the Farm, Residential, and Industrial zones as connected parts of the same distribution network. Each zone card summarizes its main measurements, operating mode, valve position, and reported online status.

This visualization provides a quick overview of the distribution network without requiring the operator to inspect every individual zone panel. The separate zone dashboards remain available for detailed monitoring and manual control.

2.7 ThingsBoard Rule Chain and Alarm Management

The ThingsBoard Root Rule Chain is responsible for processing incoming messages, evaluating monitoring conditions, and managing alarm creation and clearing. It complements the Node-RED control logic by providing a separate mechanism for detecting abnormal conditions and displaying alarms through ThingsBoard.



Message Processing and Routing

The Rule Chain starts by filtering incoming messages and checking the entity type. A message-type switch then routes messages according to their type, including telemetry, attribute updates, device activity events,

inactivity events, and RPC requests. Dedicated nodes handle telemetry and attribute storage, RPC processing, and message logging.

Water Level Monitoring

Separate script nodes evaluate low and high water-level conditions. When a configured condition is met, the corresponding Create Alarm node generates an alarm. Clear Alarm nodes handle the recovery conditions, allowing the alarm to be cleared when the water level returns to the configured acceptable range.

Flow Rate Monitoring

The Rule Chain includes separate logic for low and high flow conditions. Script nodes evaluate whether the measured flow rate meets the configured alarm conditions, while dedicated nodes create or clear the corresponding alarms. This separation allows low-flow and high-flow conditions to be managed independently.

High Temperature Monitoring

A temperature evaluation script checks incoming temperature measurements against its configured threshold. The corresponding alarm logic creates a High Temperature alarm when the condition is met and clears it when the configured recovery condition is satisfied.

Leak Detection and Sensor Faults

Dedicated script nodes evaluate suspected leak conditions and sensor faults. When their configured conditions are satisfied, the Rule Chain can create the corresponding alarms. Separate clearing logic is provided to manage recovery from these conditions.

ESP Offline Detection

The Rule Chain includes dedicated logic for detecting device inactivity and recovery. An Inactivity Event can trigger the ESP Offline alarm path, while an Activity Event follows the recovery path to evaluate and clear the alarm. This mechanism is intended to identify controllers that stop communicating and recognize when they become active again.

Alarm Management and Integration

The Rule Chain uses dedicated Create Alarm and Clear Alarm nodes to manage the lifecycle of different alarm types. These alarms complement the ThingsBoard dashboard, where operators can review system status alongside telemetry and historical charts.

The Rule Chain operates separately from Node-RED, which processes sensor measurements and executes the centralized decision logic for water distribution and valve control. Because the two systems use separate evaluation paths, their thresholds and conditions must be configured consistently and tested to ensure that their results agree.

The diagram represents the configured processing and alarm paths. Successful operation of every alarm, particularly ESP Offline, leak detection, and sensor-fault recovery, must be confirmed through runtime testing rather than inferred from the presence of nodes in the Rule Chain.

2.8 Dashboard Evaluation

The dashboard was evaluated using incoming telemetry and control commands during system testing. The monitoring panels displayed changing water-level and flow-rate readings, while the historical charts showed the corresponding trends for the individual zones.

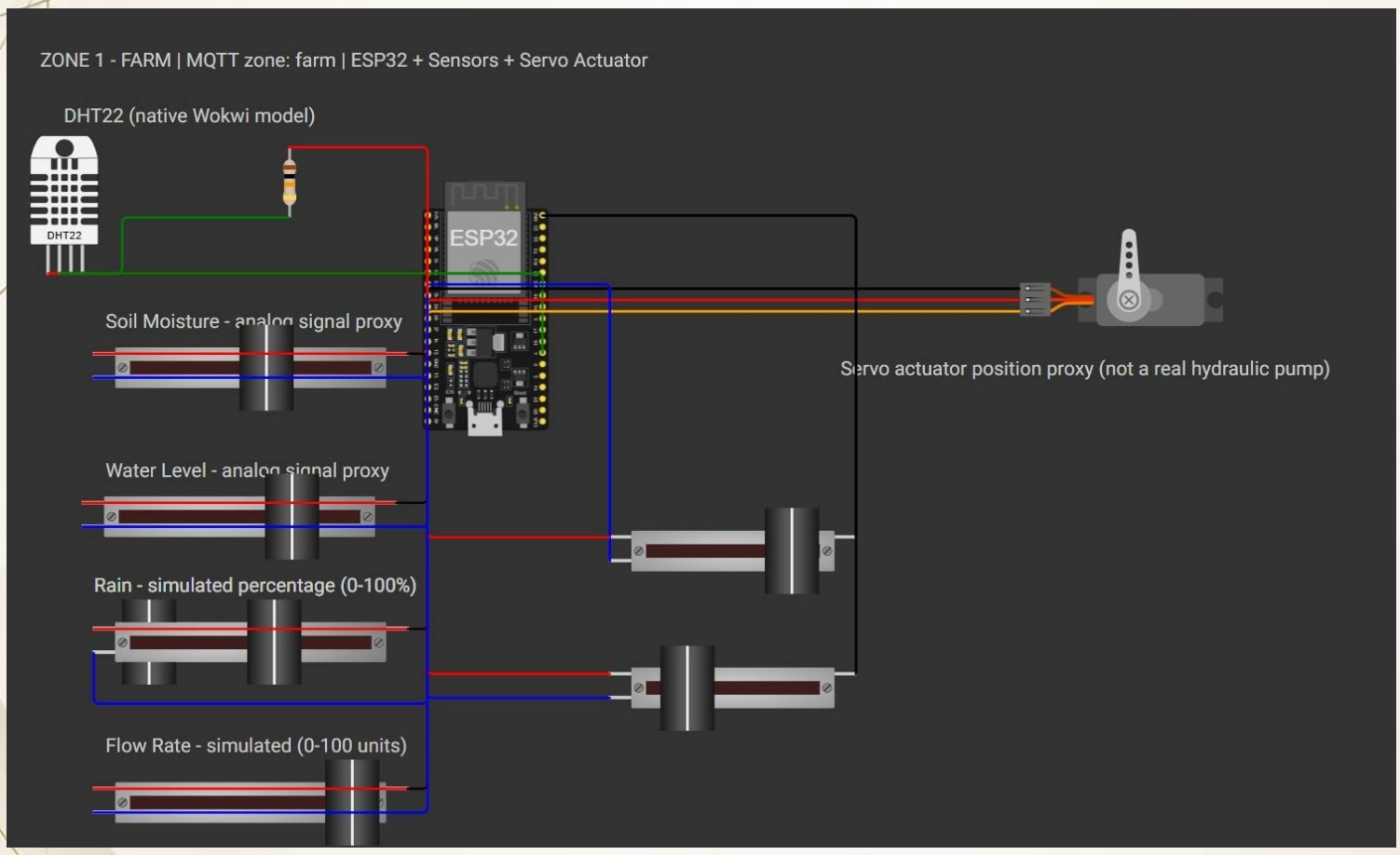
The AUTO and MANUAL controls provide two different approaches to operating the system, while the CLOSE VALVE button and valve-angle slider support direct valve control. The observed need for a page refresh during some manual valve-opening operations was documented as an area for future improvement.

Overall, the dashboard provides a unified interface for monitoring water distribution, observing historical measurements, switching operating modes, and accessing the available valve controls.

3. Simulation Design and Implementation

3.1 Farm Zone Simulation

The Farm Zone was simulated using an ESP32 in Wokwi, connected to environmental sensors, potentiometers, and a servo motor representing the irrigation valve.



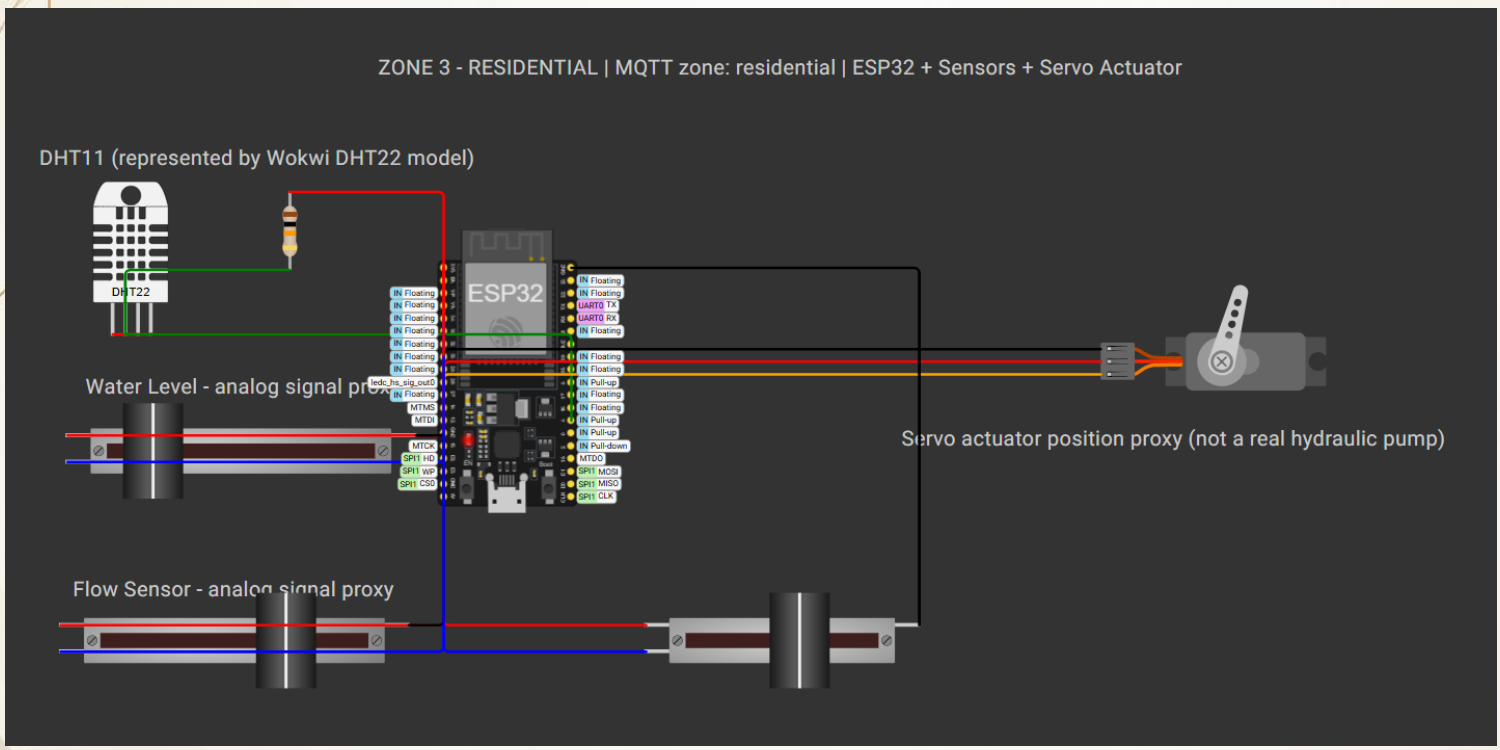
The simulated components include:

- **DHT22 Sensor:** Measures temperature and humidity to provide environmental data for the agricultural zone.
- **Soil Moisture Potentiometer:** Simulates changes in soil moisture. Adjusting the potentiometer represents different soil conditions, allowing the system to evaluate irrigation demand.
- **Water Level Potentiometer:** Simulates changes in the available water level, enabling the system to respond to different water availability conditions.
- **Rain Potentiometer:** Simulates rainfall as a percentage from 0% to 100%. This value can be used by the control logic when evaluating irrigation requirements.
- **Flow Rate Potentiometer:** Simulates the flow-rate measurement over a range of 0–100 units, allowing the system to test different water-flow conditions.
- **Servo Motor:** Represents the irrigation valve. Its angular position changes according to the control commands received by the ESP32.

These components allow the Farm Zone to simulate environmental changes, irrigation demand, water availability, and valve operation without requiring real agricultural sensors or a physical irrigation valve.

3.2 Residential Zone Simulation

The Residential Zone was implemented as a separate ESP32 circuit in Wokwi, containing a DHT sensor, two potentiometers, and a servo motor.

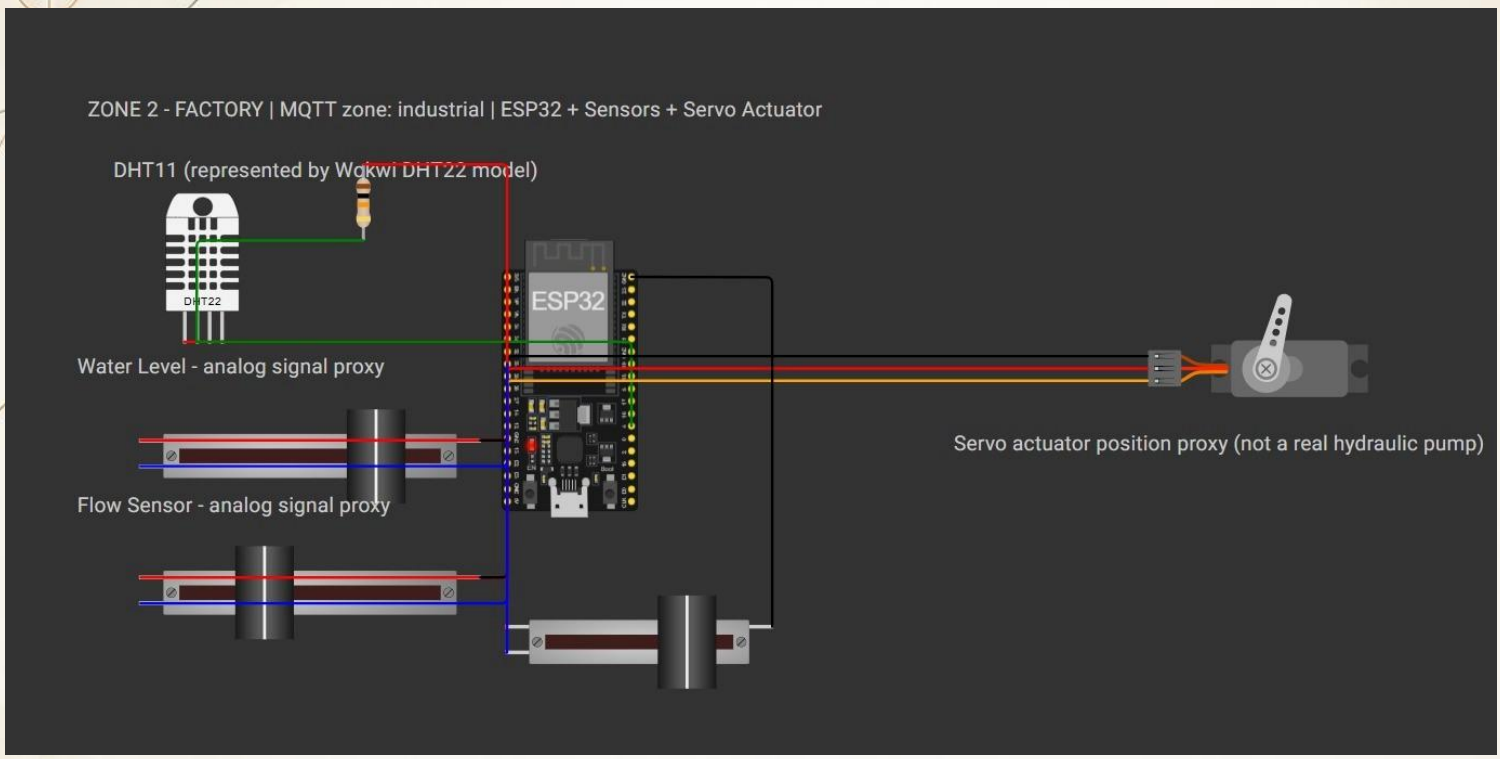


- **DHT11 Sensor:** Represents the residential temperature and humidity sensor. The DHT22 model available in Wokwi is used as a simulation substitute.
- **Water Level Potentiometer:** Simulates variations in the residential water level by providing adjustable input values.
- **Flow Rate Potentiometer:** Simulates different flow-rate readings, allowing the system to test various water-flow conditions.
- **Servo Motor:** Represents the residential distribution valve and demonstrates how its position changes in response to control commands.

The Residential Zone simulation focuses on monitoring temperature, humidity, water level, and flow rate, while demonstrating valve control under automatic and manual operating modes.

3.3 Industrial Zone Simulation

The Industrial Zone was simulated using another independent ESP32 node in Wokwi. The circuit contains a DHT sensor, analog signal proxies for water level and flow rate, and a servo actuator.



- **DHT11 Sensor:** Provides simulated temperature and humidity measurements. The Wokwi DHT22 model is used as the simulation component.
- **Water Level Potentiometer:** Represents the industrial zone's water-level measurement. Adjusting it simulates changes in the available water supply.
- **Flow Rate Potentiometer:** Represents the flow-rate input and allows different flow conditions to be tested.
- **Servo Motor:** Represents the industrial distribution valve. Its position responds to the control commands sent to the ESP32.

The Industrial Zone simulation allows the monitoring and control behavior to be evaluated independently, including changes in sensor readings and valve position.

3.4 MQTT Communication and Centralized Control

Each simulated ESP32 node represents an independent zone and publishes its sensor readings as JSON payloads through MQTT. The three zones use separate MQTT topics to identify their data:

- smartwater/farm/sensors
- smartwater/residential/sensors
- smartwater/industrial/sensors

The JSON messages contain the relevant measurements for each zone, such as temperature, water level, flow rate, and other applicable sensor values. Node-RED receives these messages, processes the incoming measurements, and applies the system's control logic.

In **AUTO mode**, Node-RED determines the appropriate valve command according to the configured control logic and operating conditions. The resulting command is sent to the corresponding ESP32, which updates the servo position.

In **MANUAL mode**, the operator can use the ThingsBoard dashboard to change the requested valve angle or issue a valve-control command. The command is passed through the control system to the corresponding ESP32, which adjusts the servo position accordingly.

This architecture separates sensor simulation, centralized decision-making, and actuator control. It also allows the three zones to be monitored and controlled independently through the same Node-RED and ThingsBoard infrastructure.

The servo motors are used as visual representations of valve positions in the simulation; they do not represent actual hydraulic pumps or demonstrate physical water flow.

4. Hardware Design and Implementation

4.1 Water Distribution System

The physical prototype consists of a main water supply line powered by a 12 V DC motor. The main line branches into three separate distribution lines, each connected to a motor responsible for transferring water to the corresponding zone reservoir.

The three zones are:

- **Farm Zone:** Supplies water to the agricultural reservoir.
- **Residential Zone:** Supplies water to the residential reservoir.
- **Industrial Zone:** Supplies water to the industrial reservoir.

Each reservoir is represented by a container in the prototype. The motors control water transfer from the main supply line to the individual reservoirs, allowing the water distribution process to be monitored and controlled for each zone.

4.2 Sensors and Zone Monitoring

Each zone is equipped with sensors to monitor its environmental conditions and water availability.

- **Temperature and Humidity Sensor (DHT):** Installed in all three zones to monitor the surrounding temperature and humidity.
- **Ultrasonic Sensor:** Installed above each reservoir to measure the water level inside the container without direct contact with the water.
- **Soil Moisture Sensor:** Installed in the Farm Zone to measure soil moisture and provide information relevant to agricultural irrigation.

The Farm Zone includes the DHT sensor, ultrasonic water-level sensor, and soil moisture sensor. The Residential and Industrial zones each include a DHT sensor and an ultrasonic water-level sensor.

4.3 Motor and Valve Control

The 12 V DC motor associated with the main supply line provides the main water supply, while the three branch motors transfer water to their respective reservoirs.

The water transfer process is controlled according to the system's operating logic. Motor operation determines whether water is supplied to a reservoir, while the control system uses the measured water level to monitor the filling process and determine the appropriate control response.

The system also supports valve-control commands for opening and closing the distribution paths where applicable. In the physical prototype, motor operation is used to control water transfer, while the simulated servo motors represent adjustable valve positions in the Wokwi environment.

4.4 Hardware Safety and Physical Separation

The prototype is arranged to keep the water containers and water-contact components physically separated from the ESP32 controllers and other electronic components.

The containers are positioned in a designated area, with sufficient separation from the control electronics to reduce the risk of water reaching the controllers. Only the required sensor wiring extends to the water-measurement points.

This arrangement helps protect the electronic components during operation and makes the prototype easier to monitor and maintain.

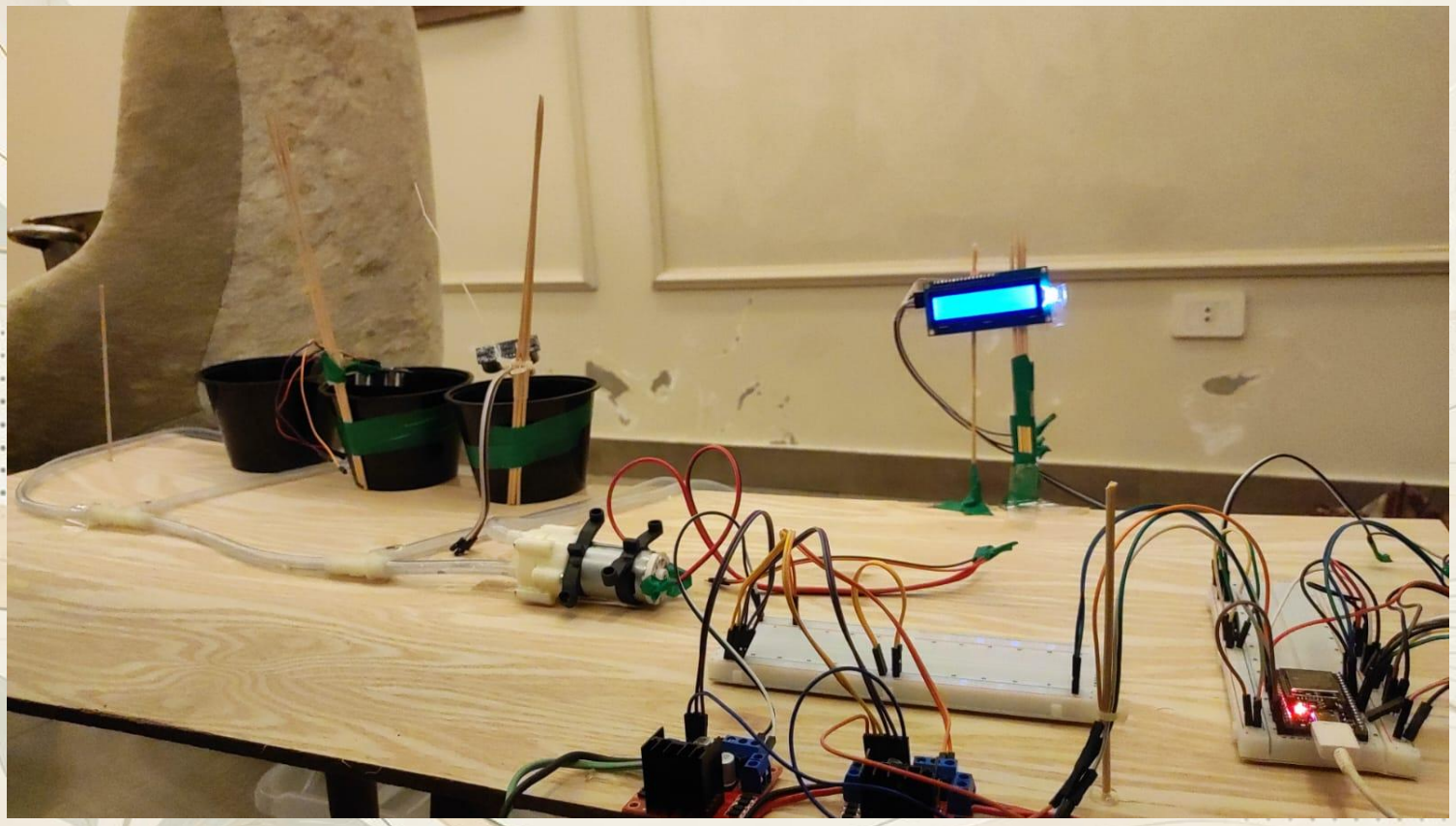
4.5 ESP32 Integration and Centralized Control

The ESP32 controllers collect the sensor readings from the physical prototype and communicate the measurements to Node-RED using MQTT messages formatted as JSON.

Node-RED processes the received measurements and applies the system's decision logic. In AUTO mode, the control logic determines the appropriate actions according to the monitored conditions. In MANUAL mode, the operator can issue control commands through the ThingsBoard dashboard.

The resulting commands are transmitted to the relevant controller, which operates the corresponding motor or control output. This allows the physical prototype to follow the same centralized monitoring and control concept used in the simulation.

The integration of the water distribution hardware, sensors, ESP32 controllers, MQTT communication, and Node-RED provides a unified approach to monitoring water levels, observing environmental conditions, and managing water distribution across the three zones.



5. Business Value, National Alignment & Validation

5.1 Target Market and Customers

The primary target customers for the Smart Water Distribution and Monitoring System are the Ministry of Water Resources and Irrigation and other governmental or national organizations responsible for water resource management and distribution.

The proposed business model is based on offering the system as a technology solution that can be implemented for public-sector customers. Services could include hardware installation, system configuration, monitoring platform integration, maintenance, and technical support.

5.2 Business Value and Expected Benefits

The system offers potential operational benefits for organizations responsible for managing water distribution:

- **Improved Water Management:** Monitoring water levels and flow conditions across multiple zones provides better visibility into water availability and distribution.
- **Reduced Water Waste:** Detecting abnormal flow conditions and suspected leaks can help operators identify potential losses and respond earlier.
- **Centralized Monitoring:** Node-RED and ThingsBoard provide a unified interface for viewing measurements, reviewing historical trends, and accessing control functions.
- **Flexible Operation:** AUTO and MANUAL modes support automated control while allowing operators to intervene when necessary.
- **Scalability:** The modular architecture can be extended to additional zones and adapted to different distribution networks.

These benefits represent the system's expected value. Actual water savings, cost reductions, and operational improvements would need to be measured during field testing.

5.3 National Alignment: Supporting Egypt's Water Management Priorities

The project aligns with broader national priorities related to water security, efficient resource utilization, digital transformation, and improved water distribution management. It is relevant to the objectives of Egypt's Water Resources Strategy 2050 and the wider framework for managing and allocating national water resources.

By providing zone-level monitoring, centralized data visualization, and controlled water distribution, the system could support operators in identifying abnormal conditions and making more informed operational decisions.

The project is not intended to solve Egypt's national water challenges independently. Instead, it addresses a specific operational need: improving the monitoring and management of water distribution across connected zones. Its potential contribution to national priorities would depend on successful field validation, integration with existing infrastructure, and evidence of measurable operational benefits.

5.4 System Validation

The prototype was evaluated through Wokwi simulation and physical hardware testing. The simulation was used to examine sensor-input variations, MQTT communication, operating modes, and servo-based valve control. The physical prototype demonstrated water transfer between the main supply line and the three zone reservoirs, together with sensor-based monitoring.

Node-RED processed the incoming measurements and applied the configured decision logic, while ThingsBoard provided live monitoring, historical charts, and access to the available control functions.

Recorded tests covered water-level conditions, abnormal flow, high temperature, sensor faults, and controller disconnection and recovery. The results demonstrated the tested monitoring functions while also identifying areas requiring further verification, including some alarm synchronization and dashboard control behavior.

5.5 Pilot and Scale-Up Plan

The next stage would be a limited field pilot conducted in cooperation with a relevant water-management authority, irrigation operator, or authorized technical partner.

Proposed pilot location: A single agricultural irrigation distribution point or small controlled irrigation zone, selected with the cooperation of the responsible operator. This would provide a manageable environment for evaluating water-level monitoring, flow behavior, and operational control before considering a larger deployment.

The pilot would follow three stages:

1. **Baseline Data Collection:** Collect measurements for approximately four weeks before deployment, where feasible, including water flow, reservoir levels, operating schedules, and relevant operational incidents.
2. **Pilot Implementation and Evaluation:** Install the system at the selected site and monitor its performance over a comparable period. Compare the results against the baseline using indicators such as abnormal-condition detection time, water losses where measurable, monitoring availability, and operator response time.
3. **Evidence-Based Scale-Up Decision:** Review the results with the operating partner. Expansion to additional zones would be considered only if the pilot demonstrates reliable operation, measurable benefits, acceptable costs, and compatibility with operational and safety requirements.

5.6 Commercial Potential

The project provides a foundation for a potential smart water management solution for governmental and national organizations. A successful pilot could support the development of a commercial offering based on system implementation, integration, maintenance, and technical support.

By validating the system in a real operating environment before expanding, the proposed approach would allow investment and scale-up decisions to be based on measured performance rather than assumptions.

6. Limitations and Future Work

Although the current system provides centralized water monitoring and control across agricultural, residential, and industrial zones, several improvements could enhance its security, predictive capabilities, and fault-detection performance.

6.1 Cybersecurity and Secure Remote Operation

Future development will focus on strengthening cybersecurity and protecting remote system operation. This includes implementing role-based access control, documenting control commands, recording important system events, and restricting unauthorized access to monitoring and control functions.

Communication security can also be improved through secure MQTT connections, appropriate credential management, and authorization rules that ensure each controller accepts commands only from trusted sources.

These improvements would help protect the system against unauthorized control commands and provide a clearer record of operational activities.

6.2 Leak Detection and Abnormality Localization

The current system can identify certain abnormal operating conditions, but future development could improve its ability to distinguish between possible leaks, blockages, and equipment faults and identify the zone most likely to be affected.

The proposed approach would compare incoming and outgoing flow measurements, changes in stored water levels, expected and measured flow conditions, and valve positions against the commands issued by the controller.

Initially, this functionality could be implemented using physical rules and statistical thresholds. As more operational data becomes available, anomaly-detection algorithms or machine-learning models could be introduced to improve fault classification and localization.

When suspicious behavior is detected, the system could notify the operator and indicate the reliability of the available measurements. Automatic isolation of an affected section could be considered only when the evidence is sufficient and the required safety conditions are satisfied.

This development could reduce fault-detection time and help minimize water losses before corrective action is taken.

6.3 AI-Based Water Demand Forecasting

Another future improvement is to introduce AI-based water demand forecasting. Instead of responding only to current conditions, the system could predict the expected water demand of each zone and adjust distribution targets in advance.

For agricultural applications, the model could use soil moisture, temperature, rainfall, and historical consumption data. For residential and industrial zones, it could use historical demand measurements and available operating schedules.

Forecasting methods such as XGBoost or time-series models could be evaluated against a simple baseline to determine whether they provide sufficiently accurate predictions. The predicted demand could then be integrated with Model Predictive Control (MPC), enabling the system to calculate suitable control actions over a future time horizon while respecting water availability and the requirements of other zones.

This improvement would move the system from purely reactive operation toward predictive water distribution, helping it prepare for changing demand and allocate available water more effectively.

Overall, these developments could transform the current prototype into a more secure, predictive, and fault-aware water management solution suitable for further evaluation in larger distribution networks.

7.Conclusion

The Smart Water Distribution and Monitoring System was developed as a prototype for monitoring and managing water distribution across three main zones: agricultural, residential, and industrial. The project combines sensor-based monitoring, embedded controllers, MQTT communication, centralized decision-making, and a web-based dashboard to demonstrate how water distribution can be managed through an integrated system.

The physical hardware prototype demonstrates the distribution of water from a main supply line to three separate reservoirs using DC motors. Sensors monitor water levels and environmental conditions, while the Farm Zone additionally includes soil moisture monitoring to support agricultural irrigation requirements. The Wokwi simulation complements the physical prototype by allowing sensor inputs, communication behavior, and servo-based valve control to be tested in a controlled environment.

ESP32 controllers communicate sensor measurements through MQTT using JSON-formatted messages. Node-RED processes the incoming data and applies the configured control logic, while ThingsBoard provides a centralized interface for viewing live measurements, reviewing historical charts, switching between AUTO and MANUAL modes, and accessing the available control functions.

Testing demonstrated the operation of several monitoring and detection functions, including water-level monitoring, abnormal-flow detection, high-temperature detection, sensor-fault handling, and controller disconnection and recovery. The recorded results also identified areas requiring further verification, particularly the synchronization of certain alarms and the behavior of some dashboard control operations.

From a business perspective, the proposed system could provide a foundation for smart water management solutions for the Ministry of Water Resources and Irrigation and other governmental organizations responsible for water distribution. Its modular design offers potential for expansion to additional zones and integration with larger monitoring networks.

Future development will focus on improving cybersecurity, enhancing leak detection and fault localization, and introducing AI-based water demand forecasting with predictive control. Further field testing, measurement validation, reliability assessment, and cost evaluation would be required before large-scale deployment.

In conclusion, the project demonstrates the potential of combining IoT technologies, centralized monitoring, and automated control to improve the visibility and management of water distribution. It establishes a foundation for developing a more secure, efficient, and scalable water management solution.